

Lily (Shi-Tao) Zhang

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Professional Summary

Master of Science in Biostatistics with a Ph.D. in Biochemistry and Molecular Biology and over six years of biomedical research experience. Skilled in applying statistical methods to clinical and population health data; developing reproducible analytical workflows; evaluating model assumptions; and communicating results to multidisciplinary research teams. Proficient in R, SAS, and Python, with experience in regression modeling, survival analysis, causal inference, and large-scale healthcare datasets.

Technical Skills

Programming: R, SAS, Python

Statistical Methods: Generalized Linear Models, Survival Analysis, Causal Inference, Propensity Score Methods, Longitudinal and Cross-sectional Data Analysis

Data Management and Reporting: Data Cleaning, Data Wrangling, Missing Data Handling, Reproducible Workflows, Quarto

Education

M.S., Biostatistics, Case Western Reserve University, Cleveland, OH GPA 4.0 Aug 2026

Ph.D., Biochemistry and Molecular Biology, Jilin University, Changchun, China Dec 2021

B.S., Life Science, Jilin University, Changchun, China June 2015

Experience

Biostatistics Intern Case Western Reserve University, Cleveland, OH May 2026–July 2026

Inflammatory Extracellular Vesicles Promote Immunosenescence and Frailty Risk in People Aging with HIV Infection

- Built reproducible analytical pipelines in R for the analysis of longitudinal and cross-sectional plasma and extracellular vesicle biomarker data.
- Applied generalized linear models, survival analysis, and multiple-testing correction to identify biomarkers associated with frailty status and progression while evaluating model assumptions and statistical validity.
- Developed publication-quality figures and reproducible analysis code to support manuscript preparation and scientific communication.
- Presented analysis progress regularly to the PI and collaborating researchers, incorporated feedback to refine analytic strategies, and translated statistical findings into biologically meaningful conclusions.

Postdoctoral Fellow Case Western Reserve University, Cleveland, OH March 2022–July 2025

Conformational Analysis of Human Estrogen Receptor α (hER α) Bound to Estrogen Response Elements (ERE)

- Designed and executed studies of hER α structure and function, using literature-driven optimization to improve experimental protocols and workflow efficiency. Increased recombinant hER α yield in *E. coli* five-fold by optimizing expression, induction, purification, and buffer conditions. Improved cryo-EM structural resolution approximately three-fold through the incorporation of protective additives during sample preparation.
- Optimized EMSA sample preparation, reagent composition, and experimental procedures through systematic troubleshooting and iterative testing, establishing a standardized and reproducible workflow that consistently generated reliable experimental results.
- Analyzed complex biological datasets, developed scientific visualizations, and contributed to manuscripts preparation and scientific presentations.

Ph.D Jilin University, Changchun, China September 2015–December 2021

- Trained and mentored junior lab members in laboratory procedures, research workflows, and safety protocols; supported day-to-day laboratory operations and coordinated project activities.
- Led and collaborated with cross-functional research teams to manage multiple projects concurrently, coordinate experimental priorities, troubleshoot challenges, and ensure timely completion of research objectives.
- Organized lab meetings, team-building activities, and social events to strengthen team communication, engagement, and collaboration.

Selected Projects

Gut Microbiome Composition and Differential Abundance Across Dietary Groups

- Independent data analysis project using R, DADA2, phyloseq, and ANCOM-BC2.
- Developed an end-to-end analytical workflow for 16S rRNA sequencing data, including quality control, sequence filtering, denoising, chimera removal, taxonomic assignment, and phylogenetic tree construction.
- Analyzed microbial community composition across dietary groups using alpha diversity, beta diversity, PCoA, PERMANOVA, and dispersion testing to evaluate within- and between-group differences.
- Conducted differential abundance analysis using ANCOM-BC2 with multiple-testing correction to identify taxa associated with dietary groups, while assessing analytical robustness and reproducibility.
- Synthesized complex sequencing and statistical results into publication-style visualizations and a technical presentation,

translating analytical findings into biologically interpretable conclusions.

Comparative Effectiveness of Diabetes Medications Using 2017–2020 NHANES Data

- CWRU Design & Analysis of Observational Studies course project.
- Used propensity score (PS) matching to compare Metformin and Sulfonylurea users and reduce treatment-selection bias.
- Built linear and logistic regression models for HbA1c-related outcomes and performed diagnostics and sensitivity analyses.

Prediction of Health Outcomes in Older Adults Using 2022 HRS Data

- CWRU Statistical Methods II course project.
- Processed large-scale survey data, constructed analysis variables, and developed regression models for self-rated health and chronic conditions.
- Evaluated model performance and translated findings into clear summaries for nontechnical audiences.

Exploration of Post-Hurricane Emergency Department Utilization Patterns Using 2018 SEDD Data

- CWRU Secondary Analysis of Large Health Care Data Bases course project.
- Processed and cleaned large-scale Florida HCUP State Emergency Department (SEDD) data and defined injury outcomes using ICD-10 diagnosis codes.
- Conducted descriptive and stratified analyses to characterize emergency department utilization and injury patterns following Hurricane Michael.
- Applied multinomial logistic regression to identify demographic and clinical factors associated with patient disposition after emergency department visits.
- Summarized findings through statistical reports and data visualizations to support interpretation of post-disaster health-care utilization.

Publications and Certificates

Publications: Zhang S. et al. *Front Chem.*, 2021; Zhang S. et al. *Molecules*, 2021; Zhang S. et al. *Oncol Lett.*, 2018; Zhang S. et al. *Arch Gynecol Obstet.*, 2016; Zhang L. et al. *RSC Adv.*, 2018 **Certificates:** SAS 9.4 Programming Fundamentals